

Md Mahbub Alam

Dhaka, Bangladesh | ✉ mahbubcse96@gmail.com | 🌐 mahbub.dev | 📄 Mahbub Alam

Summary

Passionate and detail-oriented **Software Engineer** with expertise in **full-stack development** and cloud-based VoIP systems. Proficient in **PHP (Laravel, CodeIgniter)**, **JavaScript (Node.js, React, Next.js)**, and **Python (Django, FastAPI, Deep Learning)**. Skilled in **CI/CD (GitHub Actions)**, building **scalable microservices**, and designing robust REST APIs. Experienced in implementing secure DevSecOps workflows with a strong focus on **system optimization**, **cloud engineering**, and **applied AI**.

Experience

Junior Software Engineer II

Dhaka, Bangladesh

Brotecs Technologies Ltd.

Sep 2023 – Present

- Designed and developed complex **full-stack features** using the **MERN Stack**, ensuring scalability, performance, and clean architecture.
- Integrated secure **payment gateways** and enhanced backend functionality to support seamless transactions.
- Built and optimized **backend services** with **FastAPI**, **PostgreSQL**, Prisma, and integrated them with React-based frontends.
- Developed and maintained **cloud-based VoIP systems**, improving deployment efficiency and system reliability.
- Implemented and automated **CI/CD pipelines (GitHub Actions)** and **DevSecOps workflows** for continuous and secure delivery.
- Collaborated across teams on **React.js**, **Next.js**, **PHP**, **Python**, **Mongoose**, **Prisma** environments to deliver end-to-end solutions.

Projects

- Healthcare Platform** (*MERN Stack, SSLCommerz*): Multi-role system with real-time vitals monitoring and integrated payment processing.
- IoT Management System** (*FastAPI, PostgreSQL, React*): Backend APIs and interactive dashboards for device management and real-time data visualization.
- Digital Stethoscope ECG Visualizer** (*React, D3.js, AudioContext API*): Plays and visualizes heart sounds and ECG signals in real time with dynamic waveform rendering.
- Speech Emotion Recognition** (*Python, CNN, Keras*): Deep learning model classifying Bangla speech emotions with real-time Android integration.
- Next.js E-commerce Platform** (*Next.js, TypeScript, PostgreSQL*): Full-stack e-commerce application with public pages and role-based access control for admin and users.
- Mobile Mess Management App** (*React Native, Express.js, Firebase*): Manages members, meals, and shared expenses with role-based access.
- Enterprise Academic System** (*Laravel, MySQL*): Streamlines student, faculty, and administrative workflows in an academic environment.
- E-Learning Platform** (*React, Node.js, Bootstrap*): Supports quizzes, assignments, and collaborative learning features.
- Desktop Blood Management System** (*Java Swing, SQLite*): Tracks donors, inventory, and requests efficiently.

Skills

Programming Languages	PHP, JavaScript, TypeScript, Python, Java, C, C++
Frameworks & Libraries	Laravel, CodeIgniter, Node.js, Express.js, React.js, Next.js, Bootstrap, Tailwind, Material-UI, Prisma, Mongoose
Databases	MySQL, PostgreSQL, MongoDB, SQLite, Firebase
Tools & Practices	Git, GitHub, CI/CD (GitHub Actions), DevSecOps, Microservices, REST APIs, TDD
AI/ML	Deep Learning (CNN, RNN), Speech Recognition, Speech Emotion Recognition
Languages	Bangla (Native), English (Professional)

Achievements

- Active problem solver on HackerRank, specializing in algorithms and data structures.
- Earned the JavaScript (Basic) Certificate from HackerRank.
- Earned the React (Basic) Certificate from HackerRank.

Leadership & Activities

- Organized and managed multiple university programs and technical events, enhancing coordination and team engagement.
- Coordinated an inter-software company cricket tournament, promoting teamwork and community involvement.

Education

B.Sc. in Computer Science and Engineering

2019 – 2022

Stamford University Bangladesh

CGPA: 3.70 / 4.00

Higher Secondary Certificate (Science)

2014 – 2016

BAF Shaheen College, Jessore

CGPA: 4.00 / 5.00

Secondary School Certificate (Science)

2009 – 2014

Shyamkur High School

CGPA: 4.44 / 5.00

References

Sharif Hossain

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